Reg. No. | | | | | | | | | | |

[12EC81]

# EIGHTH SEMESTER B.E UNIVERSITY EXAMINATION, JUNE - 2016

## OR AND MANAGEMENT

Time: 3 Hours

Maximum Marks:70

**Instructions:**

1. Answer any **FIVE** full Questions.
2. Missing data may be suitably assumed.

1. a. What is Operations Research? **06**

   b. Explain applications of Operations Research in Industry. **08**

2. a. Explain and illustrate graphically infeasible solution, unbounded solution and feasible solution. **06**

   b. Briefly explain the assumptions in Linear programming Model. **08**

3. Explain the following with example: **14**

   i. Slack variable

   ii. Surplus variable

4. a. What are the different variations of Transportation problem? **06**

   b. Determine the initial basic feasible solution to the following transportation problem using lowest cost entry method. **08**

| Origin | Destination | | | | Supply |
|---|---|---|---|---|---|
| | | D1 | D2 | D3 | D4 | |
| | A | 11 | 13 | 17 | 14 | 250 |
| | B | 16 | 18 | 14 | 10 | 300 |
| | C | 21 | 24 | 13 | 10 | 400 |
| | **Demand** | 200 | 225 | 275 | 250 | |

5. a. Differentiate between Transportation problem and Assignment problem. **06**

   b. Solve the following assignment problem using Hungarian assignment method. **08**

| Tasks | Subordinates | | | |
|---|---|---|---|---|
| | **I** | **II** | **III** | **IV** |
| **A** | 8 | 26 | 17 | 11 |
| **B** | 13 | 28 | 4 | 26 |
| **C** | 38 | 19 | 18 | 15 |
| **D** | 19 | 26 | 24 | 10 |

6. a. Define following: 04

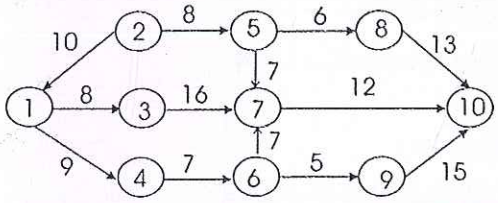
i. Network

ii. Activity

iii. Event

iv. Dangling

b. Determine the early sart ($T_E$) and late start ($T_1$) in respect of all node points and identify critical path in respect of the following network. 10




7. a. Define 06

i. Saddle point

ii. Optimal strategy and

iii. Value of game

b. What are the assumptions made in the theory of games? 08

8. a. Briefly explain the characteristics of queuing system. 06

b. In a game of matching coins with two players, suppose one player wins Rs. 2 when there are two heads and wins nothing when there are two tails; and losses Rs. 1 when there are one head and one tail. Determine the payoff matrix, the best strategies for each player and the value of the game. 08